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Global AI Trend Tracker

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EQUITY: TECHNOLOGY

AI Expert Call #70: Glass bridge and advanced packaging

Quick Note

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We hosted AI Expert Call #70 on 21 July, featuring an expert in advanced packaging for optical-electrical solutions. The expert discussed technical principles and core bottlenecks of the glass bridge solution; the impact of glass bridges on fiber array unit (FAU); edge coupling vs surface grating coupling; glass carrier vs ABF glass substrates; as well as NPO (near field packaging optics) vs CPO (co-packaged optics) and 2.5D packaging. The following are our key takeaways from the call:

Technical principles and core bottlenecks of the glass bridge solution

- Glass has good light transmittance for both visible and near-infrared light, especially near-infrared light, which is suitable for optical communication. Glass bridge is used to connect the optical chip or PIC (Photonic IC) with optical fiber, which have different mode fields. The core function is to adjust the chip's mode field to a level that matches the fiber optic mode field through the bridge.
- There are two ways to manufacture glass bridges. One is through ion exchange (sodium, silver ions), which confines light within the waveguide. This technology is relatively mature and has already been developed years ago. The second method is to use laser modification, with less efficiency and higher light loss.
- Corning (GLW US, Not rated) glass bridges use ion exchange technology.
- Current technical challenges for glass bridges include loss control, process stability and consistency, high requirements for coupling accuracy and deformation, according to the expert.

Impact of glass bridges on FAU

- Glass bridges do not completely replace FAU; instead, they add a layer of bridge structure in the middle to achieve mode-field matching and the transition from high density to sparse density.
- Glass bridges are mainly applicable in high-density optical communication scenarios.

Edge coupling vs surface grating coupling

- Edge coupling has relatively high luminous efficiency, and the coupler's optical bandwidth is larger, which is more favorable for wavelength division multiplexing and is more suitable for high-bandwidth couplers, according to the expert. Surface grating coupling has a smaller vertical grating bandwidth, so their coupling accuracy is relatively poorer. Currently, there are more edge coupling solutions being developed, according to the expert.
- The expert noted the reason why NVIDIA (NVDA US, Not rated) uses Surface coupling is that NPO/CPO requires many couplers, for example, a 6.4T CPO (at 200G per lane) needs 32 fibers, requiring very high coupling consistency. He added, edge coupling which uses 1D array cannot guarantee uniform loss control across all channels, while surface coupling which uses 2D array can have better loss control.

Glass carrier vs ABF glass substrate

- Glass carrier is essential in 2.5D packaging because many plastic encapsulation materials are used, causing shape changes and requiring glass carriers for control.
- Glass core substrate (GCS) is still far away from commercialization, although there

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are a lot of advantages such as deformation suppression, lower electrical losses, and flexibility for higher bandwidth. There are multiple technology challenges, such as low yield rate for processing TGV (through glass via), because glass is quite fragile.

- Intel (INTL US, Not rated) is a pioneer in developing GCS, and some Korean companies are also developing the solution; However, there are no reliability test data released yet. Chinese companies such as AKM Meadville (unlisted) have been sampling GCS products, and BOE (000725 CH, Buy, covered by Frank Fan) is cooperating with Corning and is currently in the development stage, according to the expert.

NPO vs CPO

- NPOs have advantages over optical transceivers in power consumption and bandwidth upgrade, according to the expert. As compared to CPO, it is easier to maintain and commercialize as it use pluggable electrical interfaces. The expert thinks the NPO market would see an earlier timeline for large-scale commercialization than CPO.

Appendix A-1

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